Problem

Let $X = \{a, b, c\}$. List all possible topologies on X. There are 29 such topologies. Compute the number of orbits under the natural action of S_3 on the set T_X of all topologies on X.

All 29 topologies

Orbit 1: Indiscrete topology

$$\{\varnothing, X\}.$$

Orbit 2: One singleton open

$$\{\varnothing, \{a\}, X\}, \qquad \{\varnothing, \{b\}, X\}, \qquad \{\varnothing, \{c\}, X\}.$$

Orbit 3: One doubleton open

$$\{\varnothing, \{a, b\}, X\}, \qquad \{\varnothing, \{a, c\}, X\}, \qquad \{\varnothing, \{b, c\}, X\}.$$

Orbit 4: One singleton and a doubleton containing it

$$\{\varnothing, \{a\}, \{a, b\}, X\}, \qquad \{\varnothing, \{a\}, \{a, c\}, X\},$$
$$\{\varnothing, \{b\}, \{a, b\}, X\}, \qquad \{\varnothing, \{b\}, \{b, c\}, X\},$$
$$\{\varnothing, \{c\}, \{a, c\}, X\}, \qquad \{\varnothing, \{c\}, \{b, c\}, X\}.$$

Orbit 5: One singleton and the complementary doubleton

$$\{\varnothing, \{a\}, \{b, c\}, X\}, \qquad \{\varnothing, \{b\}, \{a, c\}, X\}, \qquad \{\varnothing, \{c\}, \{a, b\}, X\}.$$

Orbit 6: Two singletons and their union

$$\{\varnothing, \{a\}, \{b\}, \{a, b\}, X\},$$
$$\{\varnothing, \{a\}, \{c\}, \{a, c\}, X\},$$
$$\{\varnothing, \{b\}, \{c\}, \{b, c\}, X\}.$$

Orbit 7: One singleton and two doubletons containing it

$$\{\varnothing, \{a\}, \{a, b\}, \{a, c\}, X\},$$
$$\{\varnothing, \{b\}, \{a, b\}, \{b, c\}, X\},$$
$$\{\varnothing, \{c\}, \{a, c\}, \{b, c\}, X\}.$$

Orbit 8: Two singletons and three doubletons, omitting one doubleton

$$\{\varnothing, \{a\}, \{b\}, \{a,b\}, \{a,c\}, X\},$$
$$\{\varnothing, \{a\}, \{c\}, \{a,b\}, \{a,c\}, X\},$$
$$\{\varnothing, \{a\}, \{b\}, \{a,b\}, \{b,c\}, X\},$$
$$\{\varnothing, \{b\}, \{c\}, \{a,b\}, \{b,c\}, X\},$$
$$\{\varnothing, \{a\}, \{c\}, \{a,c\}, \{b,c\}, X\},$$
$$\{\varnothing, \{b\}, \{c\}, \{a,c\}, \{b,c\}, X\}.$$

Orbit 9: Discrete topology

$$\{\varnothing, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, X\}.$$

Thus, the total number of topologies is

$$1 + 3 + 3 + 6 + 3 + 3 + 3 + 6 + 1 = 29.$$

Therefore, the number of orbits of S_3 acting on T_X is

$$\boxed{|T_X/S_3| = 9.}$$

Explanation of orbit sizes

The indiscrete topology and the discrete topology are fixed by every permutation of a, b, c. Hence each forms an orbit of size 1.

The remaining seven orbits have sizes 3 or 6, depending on whether the representative topology has nontrivial symmetry under permutations of a, b, c. Since

$$1 + 3 + 3 + 6 + 3 + 3 + 3 + 6 + 1 = 29,$$

the orbit decomposition accounts for all 29 topologies on X.

Problem

Let X be a set, and let $\mathcal{S}_1$ and $\mathcal{S}_2$ be two subbases on X, generating topologies $\mathcal{T}_1$ and $\mathcal{T}_2$, respectively.

1. If $\mathcal{S}_1 \subseteq \mathcal{S}_2$, prove that $\mathcal{T}_1 \subseteq \mathcal{T}_2$.
2. Give an explicit example where $\mathcal{T}_1 = \mathcal{T}_2$ even though $\mathcal{S}_1 \subsetneq \mathcal{S}_2$.

(i) Proof that $\mathcal{T}_1 \subseteq \mathcal{T}_2$

By definition, the topology generated by a subbasis $\mathcal{S}$ is the smallest topology on X containing $\mathcal{S}$. Equivalently, it is the collection of arbitrary unions of finite intersections of elements of $\mathcal{S}$.

Since $\mathcal{S}_1 \subseteq \mathcal{S}_2$, and since $\mathcal{T}_2$ is a topology containing $\mathcal{S}_2$, we have

$$\mathcal{S}_1 \subseteq \mathcal{S}_2 \subseteq \mathcal{T}_2.$$

Thus $\mathcal{T}_2$ is a topology on X that contains $\mathcal{S}_1$.

But $\mathcal{T}_1$ is the smallest topology on X containing $\mathcal{S}_1$. Therefore,

$$\mathcal{T}_1 \subseteq \mathcal{T}_2.$$

Alternatively, let $U \in \mathcal{T}_1$. Then U can be written as a union of finite intersections of elements of $\mathcal{S}_1$:

$$U = \bigcup_{\alpha \in A} \left(\bigcap_{i=1}^{n_\alpha} S_{\alpha i} \right),$$

where each $S_{\alpha i} \in \mathcal{S}_1$.

Since $\mathcal{S}_1 \subseteq \mathcal{S}_2$, each $S_{\alpha i} \in \mathcal{S}_2$. Hence each finite intersection

$$\bigcap_{i=1}^{n_\alpha} S_{\alpha i}$$

belongs to $\mathcal{T}_2$. Since $\mathcal{T}_2$ is closed under arbitrary unions, it follows that

$$U \in \mathcal{T}_2.$$

Therefore,

$$\boxed{\mathcal{T}_1 \subseteq \mathcal{T}_2.}$$

(ii) Example where $\mathcal{T}_1 = \mathcal{T}_2$

Let

$$X = \mathbb{R},$$

and define

$$\mathcal{S}_1 = \{(a, b) : a < b\},$$

the collection of all open intervals in $\mathbb{R}$, and

$$\mathcal{S}_2 = \{(a, b) : a < b\} \cup \{(a, \infty) : a \in \mathbb{R}\}.$$

Clearly,

$$\mathcal{S}_1 \subsetneq \mathcal{S}_2,$$

because each ray (a, ∞) is in $\mathcal{S}_2$ but is not an open interval of the form (c, d).

Let $\mathcal{T}_1$ be the topology generated by $\mathcal{S}_1$, and let $\mathcal{T}_2$ be the topology generated by $\mathcal{S}_2$.

The collection $\mathcal{S}_1$ generates the usual topology on $\mathbb{R}$. Also, every ray is a union of open intervals:

$$(a, \infty) = \bigcup_{n=1}^{\infty} (a, a+n).$$

Hence $(a, \infty) \in \mathcal{T}_1$, so

$$\mathcal{S}_2 \subseteq \mathcal{T}_1.$$

Therefore, since $\mathcal{T}_2$ is the smallest topology containing $\mathcal{S}_2$, we get

$$\mathcal{T}_2 \subseteq \mathcal{T}_1.$$

On the other hand, $\mathcal{S}_1 \subseteq \mathcal{S}_2$, so by part (i),

$$\mathcal{T}_1 \subseteq \mathcal{T}_2.$$

Combining both inclusions,

$$\boxed{\mathcal{T}_1 = \mathcal{T}_2.}$$

Thus, even though $\mathcal{S}_1 \subsetneq \mathcal{S}_2$, both subbases generate the same topology, namely the usual topology on $\mathbb{R}$.
